XP

KAIZEN XP

The MTBA Assist Tracker

Measure how often a job needs a hand.
Then prove you fixed it.

LOG IT. MEASURE IT. KILL IT.

FILLABLE ON SCREEN. PRINTABLE ON PAPER.

*"Do what nobody else can be bothered to do,
and you'll get what nobody else gets."*

FOR OPERATORS, TECHNICIANS AND SHIFT LEADERS

READ THIS FIRST

The small stops nobody logs are usually costing more than the big breakdown everyone talks about.

MTBA stands for Mean Time Between Assists. An assist is any time someone has to step in to keep a job running: clearing a jam, nudging a misfeed, resetting a fault, re-handling stock, re-doing work. Each one only takes a minute or two, which is exactly why nobody writes them down.

MTBA is how long, on average, the job runs before it needs one of those hands. The higher the number, the healthier the process. When you remove the cause of an assist, MTBA goes up, and you can prove it.

MTBA = RUN TIME ÷ NUMBER OF ASSISTS

An 8-hour shift is 480 minutes. 12 assists in that shift means MTBA = 480 ÷ 12 = 40 minutes.
Fix the main cause and get it down to 4 assists, and MTBA becomes 120 minutes: three times longer between hands.

THE FOUR STEPS

1. **Log every assist for one week**
   Use the Shift Assist Log. Every single one, however small. Page 3.
2. **Find your biggest one**
   Add up the week on the Weekly Summary. The type with the most minutes lost goes first. Page 6.
3. **Fix the cause, not the symptom**
   Use 5 Whys or the Kaizen XP RCFA Template on that one assist type.
4. **Log again and prove it**
   One more week of logging, then fill in the Before & After page. Page 7.

MTBA OR MTBF?

MTBF (Mean Time Between Failures) is for breakdowns that need an engineer. MTBA is for the small stuff the operator fixes in a minute. Track both if you can, but MTBA is where the quick wins hide.

HOW TO USE THIS PDF

On screen: open it in the free Adobe Acrobat Reader app (phone or computer), or in Chrome or Edge. Tap a box to type. Boxes with a small gold arrow open a list to pick from. Save a copy each time you start a new sheet. Some built-in phone PDF viewers do not show the lists, so use Acrobat Reader if they are missing.
On paper: print the pages you need and write straight in the boxes. The lists work as write-in boxes.

SHIFT ASSIST LOG

LOG SHEET 1

Every assist, as it happens. If you had to step in, it goes on the list.

DATE _____ **SHIFT** _____ **LINE / MACHINE / AREA** _____

LOGGED BY _____ **RUN TIME THIS SHIFT (MINS)** _____

TIME	WHAT HAPPENED	ASSIST TYPE	MINS LOST	FIXED BY
02:14	*Label web snapped at splice*	*Jam or blockage*	*6*	*Sam*

END OF SHIFT

TOTAL ASSISTS	MINUTES LOST	MTBA (MINS)	BIGGEST TYPE
count the rows	add up the column	run time ÷ assists	most common today

SHIFT ASSIST LOG

LOG SHEET 2

Every assist, as it happens. If you had to step in, it goes on the list.

DATE | SHIFT | LINE / MACHINE / AREA

LOGGED BY | RUN TIME THIS SHIFT (MINS)

TIME	WHAT HAPPENED	ASSIST TYPE	MINS LOST	FIXED BY

END OF SHIFT

TOTAL ASSISTS
count the rows

MINUTES LOST
add up the column

MTBA (MINS)
run time ÷ assists

BIGGEST TYPE
most common today

WORKED EXAMPLE: ONE WEEK ON LINE 3

What the numbers look like once you have logged a week and added it up.

Line 3 runs 15 shifts a week at 450 minutes of run time each, so 6,750 minutes a week. The team logged every assist for a full week, then added it up.

ASSIST TYPE	HOW MANY	MINUTES LOST	RANK
Jam or blockage	**58**	**348**	**1**
Misfeed or misalignment	**21**	**84**	**2**
Reset or restart	**14**	**28**	**3**
Re-handle or re-work	**6**	**30**	**4**
Manual adjustment	**9**	**18**	**5**

WHAT IT TELLS YOU

Total: 108 assists, 508 minutes lost. MTBA = 6,750 ÷ 108 = 62 minutes. Jams are more than half of all assists and two thirds of the lost time, so jams get fixed first. Everything else waits.

AFTER THE FIX

Same line, two weeks later, after the splicing standard went in.

	BEFORE	AFTER
Assists a week	**108**	**57**
MTBA	**62 mins**	**118 mins**
Minutes lost a week	**508**	**232**

THE SENTENCE

"We nearly doubled MTBA on Line 3, from 62 to 118 minutes, and got back about 4.6 hours of run time a week." That line goes in your Weekly Wins email, your improvement log and your review.

WEEKLY SUMMARY

FIND YOUR BIGGEST ONE

Add up a week of log sheets. The type with the most minutes lost is the one to kill first.

WEEK OF _____ **LINE / MACHINE / AREA** _____

ASSIST TYPE	HOW MANY	MINUTES LOST	RANK
Jam or blockage			
Misfeed or misalignment			
Reset or restart			
Manual adjustment			
Re-handle or re-work			
Material or supply issue			
Other			

THE WEEK IN NUMBERS

RUN TIME	TOTAL ASSISTS	MTBA	MINUTES LOST
minutes this week	all types	run time ÷ assists	all types

COST OF AN HOUR (£) _____ **COST THIS WEEK (£)** _____ **PER YEAR (x 52)** _____

Cost this week = minutes lost ÷ 60 x cost of an hour. Not sure what an hour costs? Use your own hourly rate as the lowest honest number, or try the Kaizen XP KPI Impact Calculator.

THE ONE WE FIX FIRST _____ **OWNER** _____

WHAT WE THINK IS CAUSING IT (THEN PROVE IT WITH 5 WHYS OR AN RCFA)

BEFORE & AFTER

PROVE IT

One week before the fix, one week after. This is the page you show people.

LINE / MACHINE / AREA _____ **ASSIST TYPE FIXED** _____

	BEFORE THE FIX	AFTER THE FIX
Week of		
Run time (minutes)		
Number of assists		
MTBA (minutes)		
Minutes lost to assists		
Cost of lost time (£)		

THE DIFFERENCE

MTBA IS NOW
times longer (after ÷ before)

MINUTES BACK
per week (before minus after)

WORTH PER YEAR (£)
cost saved per week x 52

WHAT WE CHANGED

WHO SAW IT? _____ **NAME(S)** _____

YOUR SENTENCE: "WE STRETCHED MTBA ON [LINE] FROM [X] TO [Y] MINUTES, AND GOT BACK [Z] A WEEK."

MTBA STRETCHED

Now make sure it gets seen.

The rest of the Kaizen XP range

THE VISIBILITY KIT

Six tools that make sure the work you already do actually gets seen.

THE FRONTLINE CI STARTER KIT

Five simple tools for spotting problems and making the fix stick.

THE RCFA TEMPLATE

The full root cause investigation, for faults that keep coming back.

THE KPI IMPACT CALCULATOR

Put a pound sign on any problem in two minutes.

THE BREAKDOWN ANALYSIS DASHBOARD

MTBF, MTTR and a Pareto of which machine to fix first.

A before and after like this is the strongest line on any review.

All available from Kaizen XP.

XP

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